HUY NGUYEN

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EDUCATION

University of California - Santa Barbara — Expected June 2028
B.S. Electrical Engineering

- **Honors/Awards:** Amazon Future Engineer (1/400 nationally), Elks Scholar (1/480 nationally), UCSB Promise Scholar (Full scholarship)
- **Coursework:** Data Structures & Algorithms, Database System, Signals & Systems, Digital Logic, Control System, Analog Circuit Analysis
- **Clubs**: Collegiate Tennis, Gaucho Rocket Project (Avionics Team), Ascend Consulting Group (SWE Consulting), Sigma Eta Pi

EXPERIENCE

Software Development Engineer Intern — June 2026 - September 2026
Amazon – Alexa — *Seattle, WA*

- Designed a full-stack Python WebRTC interactive agentic character calling system for Alexa+, actively requested by 15,000+ customers
- Kept conversational latency under 1.5s by solving a mismatch between speech generation and audio playback speed
- Implemented live interruption handling which cleared generated audio when callers interrupted the character mid-sentence
- Shipped the end-to-end system to Amazon's pre-production infrastructure with lip-synced 2D avatar for real calls

Software Engineer Intern – Mobile & Product — April 2026 - June 2026
Polymarket — *Remote*

- Synthesized 300+ user interviews to identify onboarding barriers in Polymarket's mobile experience, where 45% of users were not trading
- Developed a trading heat map and dual "Analyst vs. Casual" mobile experience tailored to users with different levels of trading expertise

Software Development Engineer Intern — June 2025 - September 2025
Amazon – Alexa — *Seattle, WA*

- Gave Alexa+ customers access to their call history through voice requests by integrating three backend services together for the 1st time
- Built an LLM evaluation framework for 50+ test scenarios to catch inconsistent responses & reduced AI hallucinations by context injection
- Owned end-to-end deployment of the feature and replaced a multi-step mobile app process to solve a top customer privacy complaint

PROJECTS

iGauchoBack - UC Degree Roadmap | *Next.js, TypeScript, Supabase, React Flow, Upstash, Sentry* — May 2026 - Present
igauchoback.vercel.app

- Launched a UC degree-planning platform with 70+ interactive roadmaps across multiple campuses to transform traditional advising
- Designed a pipeline that converts inconsistent UC course catalogs into validated prerequisite graphs by turning raw data into JSON schemas

Orbit AFE | *Amazon Intern Hackathon (Placed 3/100 teams)* — July 2026
orbitafe.vercel.app

- Launched Orbit in 24 hours, connecting interns across 25+ NA hubs through real-time 2D/3D presence maps with opt-in location sharing
- Architected real-time DMs, group chat, and nine-signal matching that converts and ranks potential connections from user profile data

Chromatic Guitar Tuner | *Python, SoundDevice, NumPy* — April 2026 - May 2026

- Built a real-time Python guitar tuner that analyzes live microphone audio to identify notes and determine whether they are sharp, flat, or in tune
- Designed an FFT pitch detection pipeline w/ thread-safe audio buffering to separate signal from microphone capture for responsive analysis

LiDAR Home Security Detection System | *Place Top 5 @ Northrop Grumman Expo* — September 2024 - December 2024

- Built a servo-driven TOF10120 Arduino LiDAR system scanning 180° in 1° increments to detect movement up to 170cm
- Hardened movement detection against sensor noise and failed readings using a 100cm filter, adaptive baselines, and automatic retries

LEADERSHIP

Design & Make Ambassador | *Autodesk x UCSB* — February 2026 - Present

- Selected as 1 of 2 to launch Autodesk's UCSB presence, reaching 100+ students and 4+ talks across engineering, business, and design org

Vice President | *Sigma Eta Pi - Professional Fraternity* — October 2025 - September 2026

- Founded & scaled the largest entreprenuer org on campus in two quarters; established a structured 7-week founder education program

SKILLS & INTERESTS

Languages: Java, Python, C++, TypeScript, JavaScript, SQL, MATLAB, HTML/CSS
Tools/Frameworks: AWS (EC2, S3, SSM), WebRTC, REST APIs, Spring, GStreamer, ElevenLabs, Pandas, NumPy, Maven/Gradle, CI/CD, Git
Embedded & Hardware: Arduino Uno R3, I2C, LTspice, Oscilloscope, Soldering, Digital Multimeter, Fusion 360, Onshape, Revit
Hobbies: Speed Cubing, 4x4 Rock Crawling, Sports Photography, Sticker Collecting